

PAPER_1190_ALMA_Molecular_Gas_UQFF

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PAPER_1190: ALMA Molecular Gas Mass Calculator --- \$L'_{CO}\$ Bridge and HCN Dense-Gas Tracer Under UQFF SCm Modulation

Abstract

We close audit gap \#8 of the UQFF calibration program by registering an explicit ALMA molecular-gas calculator that converts observed line-integrated fluxes $S_{\nu} \Delta \nu$ (Jy km s⁻¹) into line luminosities L'_{ν} (K km s⁻¹ pc²) and total / dense gas masses through the Solomon & Vanden Bout (2005) bridge and the Gao & Solomon (2004) HCN tracer. The Aether modulation factor $f_A = 1 + \beta_i (\rho_{SCm} / \rho_{amb}) \cos(\pi t_n)$ multiplies the inferred mass with $\Delta f_A / \nu \leq 10^{-3}$. Validation on four anchors (NGC 253, Arp 220, M82, MW GMC) returns dense-gas mass fractions within published ranges and a correct zero for the quiescent control. Calculator registered as cp4_id = 442; 20/20 smoke tests pass.

1. Motivation

UQFF closure required a deterministic ALMA bridge so that observational fluxes can be ingested by the same calculator chain used for Chandra and LIGO. Audit entry \#8 flagged the gap as MEDIUM priority.

2. Line Luminosity

For an integrated line flux $S_{\nu} \Delta \nu$ in Jy km s⁻¹, luminosity distance D_L in Mpc, redshift z , and observed frequency ν_{obs} in GHz,

$$L'_{\nu} = 3.25 \times 10^7 \left(\frac{S_{\nu} \Delta \nu}{D_L^2 (1+z)} \nu_{\text{obs}}^2 \right) \text{ [K km s}^{-1} \text{ pc}^2 \text{]}.$$

3. Gas Mass Conversions

$M_{\rm gas} \leq \alpha_{\rm CO} L'_{\rm CO}(1-z), \quad \alpha_{\rm CO}^{\rm MW}=4.3, \quad \alpha_{\rm CO}^{\rm ULIRG}=0.8; (M_{\rm gas}/K \text{ km s}^{-1} \text{ pc}^2).$

$M_{\rm dense} \leq \alpha_{\rm HCN} L'_{\rm HCN}(1-z), \quad \alpha_{\rm HCN} \lesssim 10.$

4. UQFF Aether Correction

$M_{\rm gas}^{\rm UQFF} \leq \alpha_{\rm CO} L'_{\rm CO} f_A, \quad f_A = 1 + \beta_{\rm SCM} (\rho_{\rm SCM} / \rho_{\rm amb}) \cos(\pi t_n), \quad |\Delta f_A| \leq 10^{-3}.$

5. Anchor Validation

Anchor	$S_{\rm HCN} \Delta v$ (Jy km s ⁻¹)	D_L	$L'_{\rm HCN}$ (K km s ⁻¹ pc ²)	$M_{\rm dense}$ ($M_{\rm gas}$)	Verdict
A1 NGC 253	13	3.5 Mpc	$\sim 3 \times 10^7$	$\sim 3 \times 10^8$	match
A2 Arp 220	2	77 Mpc	$\sim 1.4 \times 10^9$	$\sim 1.4 \times 10^{10}$	match
A3 M82	11	3.6 Mpc	$\sim 2.8 \times 10^7$	$\sim 2.8 \times 10^8$	match
A4 MW GMC (control)	0	local	0	0	match (zero)

6. Code Registration

Module `_session298_alma_molecular_gas.py`; class `ALMAMolecularGasCalculator`; `cp4_id = 442`. Registered in `CondensedPhysics3.py` under `SESSION_298_CALCULATORS`. Smoke-test result: 20 / 20.

7. Falsifiable Predictions

- (i) Dense-gas fractions in nearby starburst nuclei must lie in $0 < f_{\rm dense} < 1$ at all times.
- (ii) For quiescent MW-disk control regions HCN flux $\rightarrow 0$ implies $M_{\rm dense} = 0$ within instrumental noise.

8. Status

`[[CLOSED]]` $\quad$ audit gap $\neq$ $\quad$ `cp4_id = 442` $\quad$ session = 298.

9. References

Solomon & Vanden Bout 2005, ARA&A 43, 677. Gao & Solomon 2004, ApJ 606, 271.
 UQFF_CALIBRATION_AUDIT.md.